

nodal.

Overview & External Testing Brief · Oct 1, 2026 · @benji107

NODALDEX.FYI · BLOCKDAG CHAIN 1404

Nodal: overview & testing brief

Nodal is a live trade-routing layer on BlockDAG (chain 1404). It quotes every registered liquidity source and sends each swap through the one that pays you the most, for a disclosed **0.15% fee**. It is open to everyone as a **clearly labelled, unaudited beta**.

What Nodal is

Nodal is a **trade-routing layer** for BlockDAG. Instead of trading against a single exchange, you submit a swap once: Nodal's router contract asks every registered liquidity source for a quote, ranks them by what you'd actually receive after fees, and sends the trade through the best one, all in a single transaction.

It charges one disclosed fee, **0.15%** of the output, shown as its own line before you confirm and hard-capped in the contract at 1%. Today Nodal has **one live source, Reef**, which is built by the same team; that is disclosed on the site, and while Reef is the only source Nodal routes through it rather than comparing venues. The site is live at **nodaldex.fyi**.

What it can do today

Fee: 0.15% of each trade's output goes to Nodal's treasury, on top of the underlying DEX's own fee (Reef: 0.30%). The fee can only be changed by a **2-of-2 hardware-wallet multisig**, and never above 1%.

Feature	What it does
Live swap	Real quotes from NodalRouter on chain 1404 and real on-chain trades, straight from the site
Best-route selection	Quotes every active source in one call and picks the highest net output after fees
Routing via BDAG	Uses a direct pool where one exists, otherwise routes through BDAG (e.g. token → BDAG → token)
Token discovery	Lists every token with a pool on the live source, read from the chain, not a hand-kept list
Exact approvals	Tokens are approved for exactly the amount being swapped, never an unlimited allowance
Slippage protection	0.5% / 1% / 3% or custom; the contract refuses any trade that would pay out less
Price-impact warnings	Over 3% shows a warning; over 10% needs a second tap; over 30% is blocked
Safety checks before signing	Blocks trades that are over the beta limit, over your balance, or made while Nodal is paused

Feature	What it does
RPC status panel	Nine community BlockDAG endpoints checked live from your browser, with automatic failover
Bridge tab	A clearly labelled preview only; no official BlockDAG bridge exists yet

Contracts & settings

Solidity 0.8.24 · OpenZeppelin 5.0.2 · EVM Berlin

NodalRouter	Live since Sep 18, 2026 0xA06f8a856896aA1836f04F758C1E5Ac5dbe24672 · verified on explorer.bdagexplorer.com
Liquidity source	Reef, via NodalReefAdapter 0x4b60D344eDA7E3D859739B5AbC1176d756E22d56 · registered Sep 30, 2026
Owner	2-of-2 multisig (Trezor + Ledger) 0x4E2401bFD24c66166fABF9Cc5cD5B6B2c5c860fc · since block 22,931,119
Fee recipient	Treasury wallet 0x8A8F4E1d70F889C5aA2579E8ff2826e4fE8B2127
Beta limits	100 BDAG · 1,000,000 NOCAP maximum per trade, enforced by the contract
Owner powers	Pause, fee ($\leq 1\%$), limits, sources all need both hardware wallets; user funds never rest in the router between trades

Security and testing

A thorough in-house review; no independent professional audit.

Check	Result
Static analysis (Slither)	Found 3 real issues, all fixed (floating pragma, unindexed event, check-effects order in rescue functions); 6 other findings reviewed and documented as intended behaviour
Router integration tests	15/15 on a local EVM; pause and per-trade limits 7/7
Reef adapter tests	10/10 against the real Reef contracts and the real router: all trade directions, multi-hop, exact 0.15% fee, slippage, deadlines, no funds left behind
Website end-to-end test	18/18 with a headless browser and test wallet on a local chain running the real contracts: every trade direction, exact approvals, limits, balance, price impact, slippage, pause, phone layout
Live mainnet trades	0.1 BDAG → NOCAP (block 22,923,850) and 5,000 NOCAP → BDAG (block 22,929,432): paid out the quoted amount, 0.15% to the treasury
Deployed code check	NodalRouter's on-chain code matches the published source byte for byte (optimizer 200, Berlin); verified on the explorer
Front-end / contract match	Every function and event the site uses matches the deployed contract
Found and fixed	Reef renames its BDAG swap functions, so BDAG trades would have failed: solved with the adapter. Earlier: a compiler stack-too-deep error, and deployment guidance that named the wrong EVM version
Wording	Site copy corrected so it never overstates: one source disclosed as same-team, custody explained accurately, bridge marked as preview

Check	Result
Vulnerability reports	Private address for security reports: security@nodaldex.fyi

Not done yet:

- Independent audit by a professional security firm
- NodalReefAdapter can't show verified source on the explorer: its verifier doesn't handle immutables. The code matches the source apart from the stored Reef address, and that proof is published in the repo

Readiness

Now: public beta

Nodal is live on BlockDAG mainnet and open to everyone as a **clearly labelled, unaudited beta**, with per-trade limits in place. The beta banner, the terms and jurisdiction check before first use, and the FAQ all say so.

Why it is reasonable to use

- Nobody can take or freeze users' funds; trades settle in one transaction or revert
- Every admin action needs two hardware wallets
- Slippage, deadlines and limits are enforced by the contract itself

What remains a risk

- No professional audit; a bug could still cost users money
- One source today, with thin liquidity: large trades move the price
- A young chain with occasionally unreliable RPCs

For testers

What to test

Use small amounts only, on chain 1404, with MetaMask or a similar wallet, at nodaldex.fyi.

- 1 **BDAG → NOCAP:** swap 0.1 BDAG. Check the quote, fee line and minimum received, then the result.
- 2 **NOCAP → BDAG:** sell a little back. Expect two wallet prompts: an exact-amount approval, then the swap.
- 3 **Limits and warnings:** enter 150 BDAG (should be blocked) and a large NOCAP amount (price-impact warning).
- 4 **Explorer:** open "View transaction" and confirm the 0.15% fee went to the treasury.
- 5 **Phone and desktop:** use the site on both; check the RPC panel.

How to report

Include what you did, what you expected, what happened, your wallet and browser, a screenshot, and the transaction hash if there is one.

Anything that could lose funds: email security@nodaldex.fyi privately, never in public, so it can be fixed or trading paused first.

Everything else (bugs, wrong numbers, confusing wording): send it to the team, or open an issue on the project's GitHub.

As in the terms, testers must not be located in or residents of the US or UK.

Known limitations. One liquidity source (Reef) today, so there is nothing to compare yet. Pools are thin, so trades can move prices a lot. Public RPC endpoints sometimes lag or go offline. Token approvals are for the exact amount, so you approve again on each new trade (safer, one extra click). The Bridge tab is a preview.

Roadmap

Each stage unlocks the next.

- 1 **Public beta with Reef (now).** Live trading, per-trade limits, feedback gathered and fixed.
- 2 **More sources.** Each new DEX with real liquidity is added with one multisig transaction. From two sources, Nodal genuinely compares prices.
- 3 **Smarter routing.** Multi-hop across sources and splitting large trades, once there is more than one venue to route across.
- 4 **Security hardening.** Independent audit with the report published; a bug-bounty programme; limits raised as the track record grows.
- 5 **Bridged assets.** The Bridge tab goes live only once an official, audited BlockDAG bridge exists.

Open question for the team: target date and budget for the independent audit.